Interconnect Design and Technology Optimization for Conventional and Emerging Nanoscale Devices: A Physical Design Perspective

(*Invited*)

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***Abstract*—**Interconnect parasitics severely limit the performance and power dissipation in modern circuits at the advanced process technology nodes. Hence, device-level advances must be complemented with appropriate interconnect technology and design innovations for effective enablement at the circuit and system levels. This paper highlights the impact of device technologies on the optimal interconnect design and circuit-level metrics. The FinFET and Tunnel-FETs are studied by building fully placed-and-routed physical designs. The impact of device and interconnect technology co-optimization on circuit performance, power, and variability is shown for a range of emerging devices.

I. Introduction

The evolution of technology, microarchitecture and design practices with technology scaling has introduced several challenges and opportunities for innovation in modern VLSI design. As the device and wire dimensions are scaled aggressively in the advanced process nodes, maintaining reliable device functionality and managing wire parasitics with efficient design implementation have caused the system-level performance to almost plateau. Copper interconnect resistivity has increased rapidly beyond the 14/16 nm technology node due to size effects, leading to the exploration of alternate materials for tight-pitch metal levels in the Back-End-Of-Line (BEOL) stack, like cobalt and ruthenium, which have less pronounced size effects [1]. On the device front, new exotic transistors have been proposed catering to diverse applications introducing new challenges in various facets of VLSI design to achieve the target Performance-Power-Area (PPA) [2]. With the increasing importance of interconnects in determining the technology scaling benefits, we argue the need for evaluating the device and interconnect interactions, and co-optimizing interconnect and device technologies to extend the Moore's scaling paradigm.

II. Optimal Interconnect Design for High-Performance FinFET

The FinFET transistor that is designed for enhanced electrostatic control of the channel exhibits lower resistance and increased device capacitance due to its non-planar structure in comparison to planar technology. This has exacerbated the interconnect resistance problem, particularly at lower metal levels in the BEOL stack [3]. The increased device capacitance amplifies the importance of wire resistance in circuit timing. Similarly, the reduced device resistance has alleviated the influence of wire capacitance on circuit timing [3]. In older technologies, prior to the onset of copper size effects, the interconnect design at the local metal levels was primarily focused on minimizing the wire capacitance while ensuring sufficient resistance to electromigration [4, 5]. The FinFET device Resistance-Capacitance (RC) combined with severe copper size effects have weighed the wire resistance to be far more important to circuit performance, while reducing the impact of wire capacitance at tight-pitch metal levels [3]. The shift in the interconnect design paradigm in the FinFET technology is demonstrated by studying fully placed-and-routed designs at the 7nm technology node (Tables I-III) [3]. Traditional interconnect geometry (Case I in Table III) is defined with equal interconnect width and spacings [3]. The definition of wire geometry is typically a tradeoff between interconnect parasitics, reliability, and manufacturability. However, to take advantage of the shift in the interconnect trends in the FinFET era, an alternate wire geometry is explored that alleviates wire resistance at the cost of larger wire capacitance (Fig. 2), primarily with the goal of improving performance [3]. Lowering the wire resistance increases the circuit performance by up to 1.5× (Fig. 3). In addition, the buffering required to meet the circuit timing is reduced with lower wire resistance; hence, lowering the overall power dissipation despite an increased wire capacitance [3].

Multi-patterning regimes using immersion lithography and EUV technology are employed to pattern narrow wires in the nano-scale regime which have rapidly increased the design rules making standard-cell design extremely difficult [1] Furthermore, variation in interconnect dimensions critically affects the system-level metrics, particularly due to the sensitivity of copper wire resistance to the wire dimensions [6]. In addition to the wire resistance problem that is worsened by the higher FinFET capacitance, it is also found that wire resistance variability is larger than wire capacitance variability [6]. Therefore, the proposed wire geometry with reduced wire resistance and higher wire capacitance (in Fig. 2-b) alleviates the impact of interconnect variability on circuit timing, in addition to improving the circuit timing at a target yield. This is studied by inducing lithography-dependent statistical interconnect variability in placed-and-routed designs (Fig. 3) [6]. A proposed novel hybrid quadruple-patterning lithography

regime (LELE+SADP) can potentially lower wire resistance variation, improve the circuit performance at a target yield, and substantially improve the variation in timing (3σ) [6, 7].

III. Interconnect Design For Low-Power Devices

In contrast to the high-performance multi-gate transistors, the low-power transistors are characterized by low gate capacitance and the capability of operating at lower supply voltages. For example, the Tunnel-FET (TFET) devices have been widely investigated thanks to their potential steep subthreshold swing (SS). However, these devices exhibit substantially higher device resistances compared to thermionic FETs [8-10]. This increased device resistance amplifies the importance of wire capacitance in circuit delay, whereas the reduced device capacitance alleviates the influence of wire resistance on circuit delay [11]. This is validated by building physical designs based on vertical-TFET (vTFET) technology libraries (Fig. 5), and using an alternate wire geometry that alleviates wire capacitance at the cost of wire resistance (Fig. 6). It is demonstrated that the energy-delay product (EDP) is improved despite worsening the wire resistance with the proposed wire geometry at aggressively scaled dimensions (Fig. 7). Furthermore, Fig. 8 shows that, both, the performance and power improve with the proposed wire geometry (Case 2), beyond which, the performance and power begin to degrade because of the excessive wire resistance (Cases 3-5).

To evaluate the sensitivity of the TFET-based circuits to wire resistance, the copper wires are replaced with various tungsten wire configurations (Table IV). Tungsten is particularly chosen as it is more resistive than copper, but exhibits superior tolerance to electromigration at narrow wire dimensions [12]. The sheet resistance is computed in Fig. 9 for copper and tungsten interconnects with varied barrier liner thickness assumptions as tungsten-based wires do not demand barrier liners as thick as those needed for copper wires [3, 12]. It is illustrated in Fig. 10 that vTFET-based physical designs can tolerate higher resistance of the tungsten wires without taking a hit on circuit performance and power. This study showcases the opportunity to replace copper with higher resistance materials that exhibit superior reliability properties or those that are easier to fabricate, broadening the scope and advantage areas for ultra-low power devices. Furthermore, interconnect variation critically impacts FinFET-based circuits [6]. However, the TFET-based circuits are less sensitive to interconnect resistance variation, and can tolerate up to 2× variation in wire resistance (see Fig. 9 and Fig. 10).

IV. Interconnect Design Techniques for Novel Device Architectures

The potential (a priori) performances of a diverse set of emerging device proposals have been evaluated in [2, 17], and some of them are listed in Table V. These devices have unique implications on the interconnect design paradigm due to their diverse RC characteristics. The device RC values presented in Table V take parasitic RC into account, which play a pivotal role in the device performance. Based on a Fanout-Of-4 (FO4) circuit setup, the optimal interconnect geometry is obtained for each device technology to achieve maximum performance (Fig. 11). The spectrum of devices in Fig. 11, ranging from more resistive to more capacitance dominated require the wire geometry to be either optimized for wire capacitance or wire resistance, respectively, to achieve optimal performance. This trend is delineated in Fig. 11, showing the unique implications of device innovations on interconnect technology and design.

V. Need for Accurate Wire Length Modeling

In recent years, accurate benchmarking of technology has gained substantial importance as a diverse set of potential technology options is being pursued. New devices are required to be evaluated in the context of large systems and demand accurate representation of interconnection networks. In prior sections, interconnects are modeled/optimized either based on fully placed-and-routed designs which are very time intensive and not always a plausible option, or they are based on simplistic FO4 circuits. However, in real systems, there are various circuit topologies which comprise unique device and wire configurations that simplistic representations fail to holistically capture. A popular method, based on Rent's rule (a relationship between number of gates and connections), has been employed in the past few decades to model the interconnect lengths in various design families [13]. This approach provides the capability to extrapolate the potential performance of a device in a large system. However, it has been shown that these methods are highly erroneous (~90% error) in projecting performance due to underestimation of the interconnect length in modern designs [13]. A modernized Rent's rule proposed in [13], aims to capture the critical facets of modern VLSI design which include standard-cell design, microarchitecture and the Electronic-Design-Automation tools which determine the design PPA [14]. The new Rent-based model facilitates accurate representation of wirelengths and is validated against a suite of state-of-the-art commercial designs across technology nodes (Fig. 12). These models substantially improve the existing benchmarking methodologies and facilitate futuristic technology pathfinding [14].

VI. Conclusions

In this paper, the implications of device characteristics on interconnect technology, design, and optimization is discussed, providing a holistic view of the impact of device innovation on interconnects, PPA, and reliability. Adapting interconnects to suit the device properties further enhances the device benefit at the circuit and system level. The FinFET-based designs are more sensitive to the interconnect resistance than the interconnect capacitance. However, low-power devices like TFET, reduce the impact of the wire resistance on the circuit and can tolerate higher variability in wire resistance, even at the nanoscale regime. Capturing the device and wire interactions, early co-optimization, and accurate wire length modeling is required for future technology pathfinding.

Acknledgements

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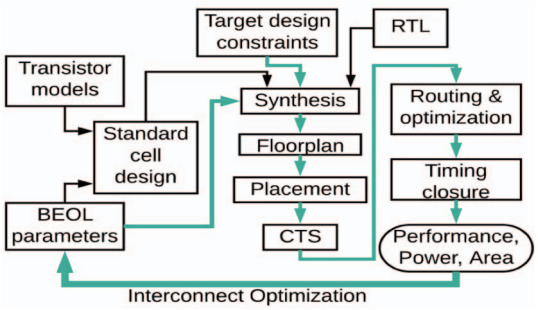


Fig. 1. VLSI-design flow utilized in this study to build GDSII-level layouts and optimize interconnects.

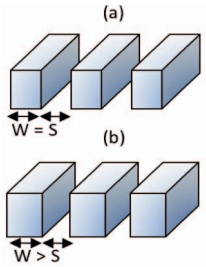


Fig. 2. (a) Wire geometry defined by the ITRS, and (b) optimized geometry proposed for high-performance FinFET designs at a constant wire pitch.

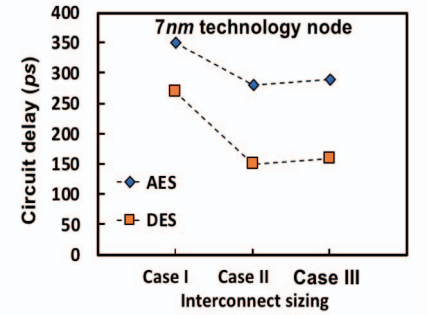


Fig. 3. Circuit delay obtained from fully placed-and-routed designs (AES and DES circuit blocks), for three cases of wire geometry.

Table I. BEOL stack assumptions at the 11- and 7-*nm* technology nodes.

11nm technology node			
Metal-levels	Metal width (W) in *nm*	Metal thickness (T) in *nm*	Metal pitch (P) in *nm*
M1-M3	17.5	35	35
M4-M6	35	70	70
7nm technology node			
M1-M3	10.8	21.8	21.8
M4-M6	21.8	43.6	43.6

Table II. FinFET Technology assumptions at the 11- and 7-nm nodes. Device models are obtained from [14].

Parameters	11nm node	7nm node
ITRS node	2018	2022
VDD	0.75	0.7
Transistor channel length	14	11

Table III. Interconnect geometries studied for FinFET-based circuits. The metal pitch (P), and wire thickness to width aspect ratio (AR), are maintained constant for the three cases (Case I-III).

Interconnect sizing at constant pitch	Wire width (W)	Wire spacing (S)
Case I	W	S=W
Case II	1.1xW	0.9xS
Case III	1.15xW	0.85xS

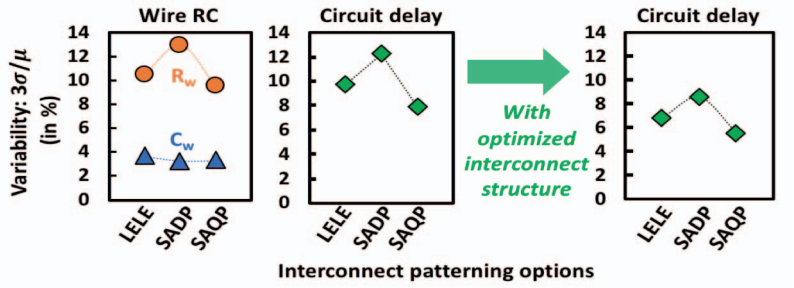


Fig. 4. Interconnect RC variability and its impact on circuit delay variability using the ITRS defined wire geometry, versus the optimized interconnect geometry (Case II on the right), in the 11*nm* technology node. The variability in circuit timing is studied on fully placed-and-routed designs for Litho-Etch-Litho-Etch (LELE) and Self-Aligned-Double/Quadruple-Patterning (SADP/QP).

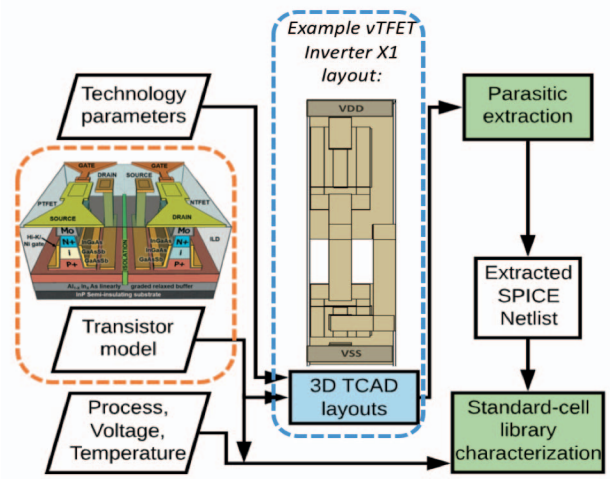


Fig. 5. Standard-cell library characterization methodology for vTFET-based layouts. The parasitic extraction is done using a 3D TCAD tool [15], and the library characterization is done using Silicon Smart Synopsys tool. The vTFET image is taken from [9].

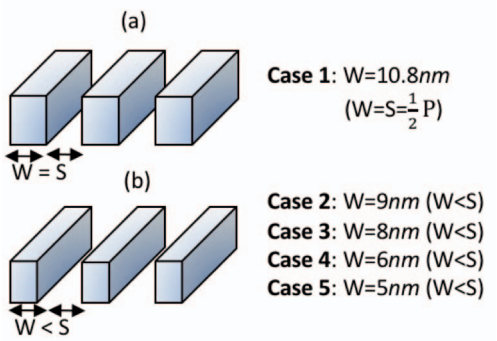


Fig. 6. (a) Wire geometry defined by ITRS, and (b) optimized geometry (Case 1), and (b) optimized wire geometry proposed for low-power vTFET-based designs, at constant P and AR (Cases 2-5).

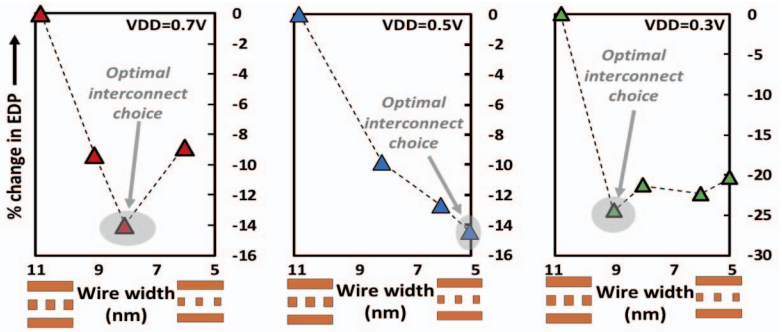


Fig. 7. EDP of a place-and-routed design (AES circuit block) based on vTFET libraries using the different interconnect geometries (Cases 1-5 in Fig. 6).

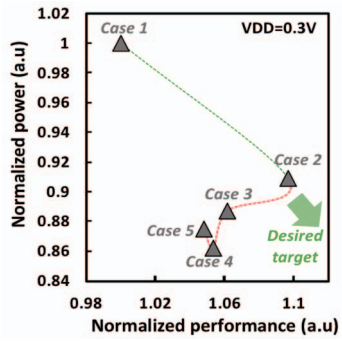


Fig. 8. Performance versus power plot for vTFET-based physical design (AES), for the wire geometries defined in Fig. 6 (Case 1-5).

Table IV. Interconnect assumptions to study the impact on vTFET-based placed and routed designs. Here, λ is the mean free path, and ρ is the bulk resistivity.

Wire composition	Wire material	Metal width (M1-M3)	Barrier thickness	(specularity *p*, reflectivity *R*)	λ	ρ
①	Copper	8*nm*	**TB1**: 1.8*nm*	(0, 0.43)	40nm	1.8μΩ-cm
②	Tungsten	8*nm*	**TB1**: 1.8*nm*	(0.3, 0.25)	28nm	8.7μΩ-cm
③	Tungsten	8*nm*	**TB2**: 1*nm*	(0.3, 0.25)	28nm	8.7μΩ-cm
④	Tungsten	8*nm*	**TB3**: 0.5*nm*	(0.3, 0.25)	28nm	8.7μΩ-cm

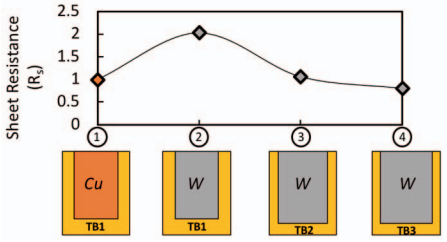


Fig. 9. Impact of interconnect setup on the sheet resistance of the interconnect (in a.u).

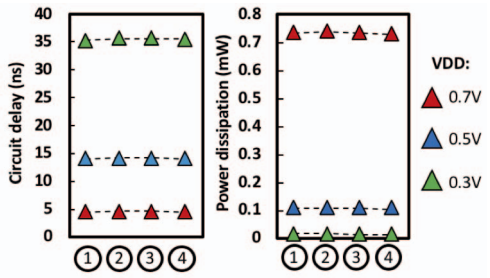


Fig. 10. Impact of interconnect material and barrier liner thickness on vTFET-based physical design timing and power.

Table V. List of the exotic devices considered in this paper. Device parameters are taken from [17] and parasitic capacitances are estimated by Raphael [16].

Device type	Naming convention	VDD (V)	Device resistance (R_o) in kΩ	Device capacitance (C_o) in aF
Planar high-performance CMOS	CMOS HP	0.73	9.4	61.4
Planar low-power CMOS	CMOS LV	0.3	97	40.3
Homo-junction III-V TFET	HomJTFET	0.2	139	66.6
Hetero-junction III-V TFET	HetJTFET	0.4	18.7	66.6
GaN TFET	GaNTFET	0.4	6.15	57.2
Transition Metal Dihalchogenide TFET	TMDTFET	0.2	46.5	76.5
Van der Walls FET (Black Phosphorous)	vdWFET-BP	0.3	148.4	39.3

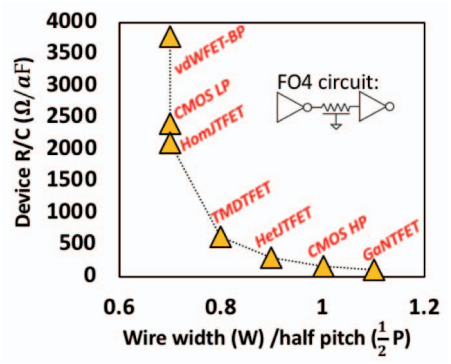


Fig. 11. Optimal interconnect geometry obtained by modifying the wire width and spacing, for a spectrum of exotic devices on the benchmarking landscape. The metal pitch and aspect ratio are maintained constant and the interconnects is assumed to be 5μm long.

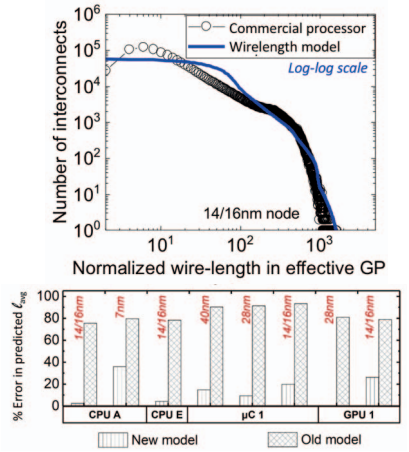


Fig. 12. Wirelength modeling: Histogram of wirelengths obtained from the model versus processor data (top), and accuracy of average wirelength projection (bottom) [14].

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